

MA1513 Exam (Exemplify)

AY2023/2024 Semester II

MCQ/FIB (15 questions; 3 marks each)

1. Given a 3 x 4 linear system $\mathbf{Ax} = \mathbf{b}$ with a general solution $\begin{cases} x_1 = s + t \\ x_2 = 3t - 4s \\ x_3 = s + 1 \\ x_4 = t - 2 \end{cases}$. Which of the

following is a particular solution of the homogeneous system $\mathbf{Ax} = \mathbf{0}$? (Pick all correct options)

a. $\begin{pmatrix} 0 \\ 0 \\ 1 \\ -2 \end{pmatrix}$

b. $\begin{pmatrix} 1 \\ -4 \\ 1 \\ -2 \end{pmatrix}$

c. $\begin{pmatrix} 1 \\ 3 \\ 0 \\ 1 \end{pmatrix}$

d. $\begin{pmatrix} 1 \\ 10 \\ 0 \\ 0 \end{pmatrix}$

e. $\begin{pmatrix} 2 \\ -1 \\ 1 \\ 1 \end{pmatrix}$

2. Suppose $\mathbf{A} = (\mathbf{c}_1 \ \mathbf{c}_2 \ \mathbf{c}_3)$ is a non-singular 3x3 matrix, where $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$ are the three columns of

A. Which of the following statements are true? (Pick all correct options)

- a. The three vectors $\mathbf{c}_1, \mathbf{c}_2$ and $\mathbf{c}_3$ are coplanar in the XYZ-space.
- b. Any row echelon form of $\mathbf{A}$ has three pivot columns.
- c. The three rows of $\mathbf{A}$ form a basis for $\mathbb{R}^3$.
- d. $\det \mathbf{A}$ is not zero.
- e. The non-homogenous system $\mathbf{Bx} = \mathbf{c}_3$ is consistent where $\mathbf{B}$ is the 3x2 matrix $(\mathbf{c}_1 \ \mathbf{c}_2)$.

3. Which of the following have the same value as the 3x3 determinant $\begin{vmatrix} a & b & c \\ d & e & 0 \\ f & 0 & 0 \end{vmatrix}$ for any

values of a, b, c, d, e and f? (Pick all correct options)

a. $\begin{vmatrix} f & 0 & 0 \\ d & e & 0 \\ a & b & c \end{vmatrix}$

b. $\begin{vmatrix} 0 & 0 & f \\ 0 & e & d \\ c & b & a \end{vmatrix}$

c. $\begin{vmatrix} -c & -b & -a \\ 0 & -e & -d \\ 0 & 0 & -f \end{vmatrix}$

d. $\begin{vmatrix} e & d & 0 \\ b & a & c \\ 0 & f & 0 \end{vmatrix}$

e. $\begin{vmatrix} 0 & 0 & f \\ c & b & a \\ 0 & e & d \end{vmatrix}$

4. Which of the following vectors belong to the row space of $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \\ 3 & 5 & 7 & 9 \end{pmatrix}$? (Pick all correct answers.)

- a. (3, 6, 9, 12)
 b. (4, 3, 2, 1)
 c. (1, 2, 4, 8)
 d. (3, 7, 5, 9)
 e. (19, 29, 39, 49)

5. The set $\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$ is a basis for the column space of which of the following matrices? (Pick all correct answers)

- a. $\begin{pmatrix} 2 & 0 & 0 \\ 2 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}$
 b. $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix}$
 c. $\begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 2 \\ 0 & 1 & 3 \end{pmatrix}$
 d. $\begin{pmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$
 e. $\begin{pmatrix} 1 & -1 & 1 & -1 \\ 1 & -1 & 1 & -1 \\ 0 & 0 & 1 & -1 \end{pmatrix}$

6. The projection $\mathbf{p}$ of the vector $\begin{pmatrix} 6 \\ 0 \\ 12 \end{pmatrix}$ onto the column space of the matrix $\begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 2 \end{pmatrix}$ is given by

$\mathbf{p} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ where $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$, $c = \underline{\hspace{1cm}}$ (FIB)

7. The coordinate vector of a vector $\mathbf{v}$ with respect to the basis $S = \{ (1, 0, 0), (0, 1, 1), (0, -1, 1) \}$ is given by $(\mathbf{v})_S = (2, 3, 5)$. Then the coordinate vector of $\mathbf{v}$ with respect to the basis $T = \{ (1, -1, 0), (1, 1, 0), (0, 0, 1) \}$ is given by $(\mathbf{v})_T = (a, b, c)$ where $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$, $c = \underline{\hspace{1cm}}$ (FIB)

8. Which of the following subspaces of $\mathbf{R}^3$ are represented by the plane with equation $3x + y - 2z = 0$ in the XYZ-space? (Pick all correct options)

- a. $\text{Span}\{ (3, 1, -2) \}$
 b. Row space of $\begin{pmatrix} 1 & -1 & 1 \\ 0 & 2 & 1 \\ 3 & 1 & -2 \end{pmatrix}$

- c. Nullspace of $\begin{pmatrix} 3 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix}$
- d. Column space of $\begin{pmatrix} 2 & 1 & 0 \\ 0 & -3 & 2 \\ 3 & 0 & 1 \end{pmatrix}$
- e. Solution space of $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

9. Let $T: \mathbf{R}^3 \rightarrow \mathbf{R}^3$ be a linear transformation such that $T\left(\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$ and $T\left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$.

Then $T\left(\begin{pmatrix} 7 \\ 10 \\ 4 \end{pmatrix}\right) = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ where $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$, $c = \underline{\hspace{1cm}}$ (FIB)

10. Given $\mathbf{A} = \begin{pmatrix} 2 & 0 & 0 & 0 \\ * & 2 & 0 & 0 \\ * & * & 3 & 0 \\ * & * & * & 3 \end{pmatrix}$ is a diagonalizable matrix, where the * can be any random numbers.

Which of the following statements are correct? (Pick all correct answers)

- a. The characteristic polynomial of $\mathbf{A}$ is $(x + 2)^2(x + 3)^2$
- b. It is possible that one of the eigenspaces of $\mathbf{A}$ has dimension 1.

c. $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 3 \end{pmatrix}$ is an eigenvector of $\mathbf{A}$

d. There is a matrix $\mathbf{P}$ such that $\mathbf{PAP}^{-1} = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$

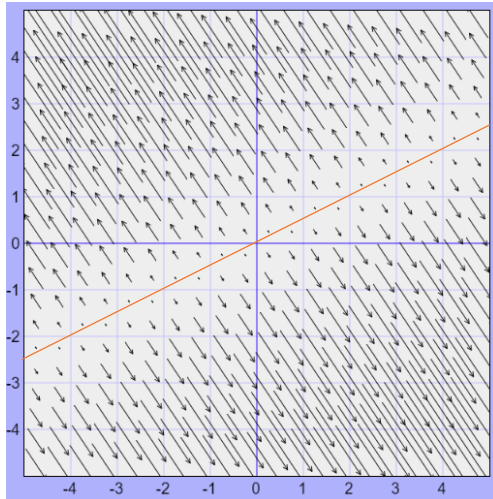
e. $\mathbf{A}^n$ is a diagonalizable matrix for any positive integer n .

11. Given a 2×2 matrix $\mathbf{A}$ with nullspace given by $\text{span}\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\}$ and $\mathbf{A}^3\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 8 \\ 16 \end{pmatrix}$. Then $\mathbf{A}$ is given by $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ where $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$, $c = \underline{\hspace{1cm}}$, $d = \underline{\hspace{1cm}}$ (FIB).

12. Given the SDE $\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$, which of the followings are solutions of the SDE? (Pick all correct answers)

- a. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} e^{3t} \sin 2t \\ e^{3t} \cos 2t \end{pmatrix}$
- b. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} e^{3t} \sin 2t \\ -e^{3t} \cos 2t \end{pmatrix}$
- c. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} e^{3t} \cos 2t \\ e^{3t} \sin 2t \end{pmatrix}$
- d. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} -e^{3t} \cos 2t \\ e^{3t} \sin 2t \end{pmatrix}$
- e. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} e^{3t} (\cos 2t + \sin 2t) \\ e^{3t} (\cos 2t - \sin 2t) \end{pmatrix}$

13. The direction field of a 2 by 2 SDE $\mathbf{y}' = \mathbf{A}\mathbf{y}$ is given below.



Which of the following statements are correct? (Pick all correct answers)

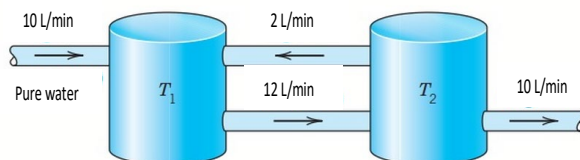
- a. The equilibrium point of the SDE is a star node
- b. There are infinitely many equilibrium points for the SDE
- c. $\mathbf{A}$ has only 1 linearly independent eigenvector
- d. $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ is an eigenvector of $\mathbf{A}$
- e. $\mathbf{A}$ is a singular matrix

14. A mass-spring system is given by the ODE: $m\mathbf{y}'' + c\mathbf{y}' + k\mathbf{y} = 0$

where k is the spring constant, c is the damping constant, and m is the mass of the ball. The ODE can be converted to an SDE $\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -k/m & -c/m \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$. Suppose the equilibrium point of the SDE is a degenerate node. Which of the following statements are correct? (Pick all correct answers)

- a. The equilibrium point is unstable.
- b. The ball will come to a halt as slow as possible without oscillation.
- c. $c/2 = \sqrt{mk}$
- d. The damping force in the system is equal to the spring force.
- e. None of the above

15. Consider the two tanks in the diagram. Tank 1 holds 100 litre of salt water and tank 2 holds 50 litre of salt water.



Suppose the amount of salt in tank 1 and 2 are $y_1(t)$ and $y_2(t)$ respectively. Then the SDE that

describe the amount of salt in the two tanks are given by $\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \frac{1}{100} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$

where $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$, $c = \underline{\hspace{1cm}}$, $d = \underline{\hspace{1cm}}$ (FIB).

Long Questions (4 questions; total 45 marks)

Q16 (10 marks)

Answer this question by writing in the exam answer booklet. Do not type in Exemplify directly. Indicate Q16 at the top on each page of the answer to this question. Indicate the part numbers on the left margin. You may use MATLAB for this question. But you should show how your answers are derived.

Given three planes P_1, P_2, P_3 on the XYZ-space with equations $x + y - z = 0$, $2x - y + z = 0$ and $x + 2y - 2z = 0$ respectively.

- (i) [3 marks] Set up a 3×3 homogeneous system with the three equations and solve the system.
- (ii) [3 marks] Describe the intersection of the three planes geometrically and express this intersection in terms of a linear span.
- (iii) [4 marks] The planes with equations $x + y - z = a$, $2x - y + z = b$ and $x + 2y - 2z = c$ are parallel to P_1, P_2, P_3 respectively, where a, b, c are some numbers, not all zero. Write down an expression of c in terms of a and b so that the three new planes have common intersect. Elaborate how you derive your answer.

Q17 (10 marks)

Answer this question by writing in the exam answer booklet. Do not type in Exemplify directly. Indicate Q17 at the top on each page of the answer to this question. Indicate the part numbers on the left margin. You may use MATLAB for this question. But you should show how your answers are derived.

Two quantities X and Y are related by a quadratic equation:

$$X^2 + aX + b = Y \quad (*)$$

for some a and b . Four sets of measurements for (X, Y) are observed:

$$(2, 10), (3, 16), (4, 25), (5, 36)$$

- (i) [2 marks] Use the measurement to set up a linear system $\mathbf{Ax} = \mathbf{B}$ in variables a and b .
- (ii) [5 marks] Find the least squares solutions of the above system, and write down the quadratic equation $(*)$ that best fits the given data. Show your workings clearly. (You may use MATLAB to compute the standard matrix operations.)
- (iii) [3 marks] For the matrix $\mathbf{A}$ and column vector $\mathbf{B}$ in (i), find the shortest possible distance between $\mathbf{B}$ and a vector in the column space of $\mathbf{A}$. Give your answer up to 4 decimal places. Explain how you derive your answer.

Q18 (12 marks)

Answer this question by writing in the exam answer booklet. Do not type in Exemplify directly. Indicate Q18 at the top on each page of the answer to this question. Indicate the part numbers on the left margin. You may use MATLAB for this question. But you should show how your answers are derived.

The annual migration of populations between town A and town B are given by the following model: Each year, 15% of town A population moves to town B, and 10% of town B population moves to town A.

- (i) [3 marks] Let a_n and b_n denote the populations in town A and B respectively at the end of year n . Find the matrix $\mathbf{A}$ such that $\begin{pmatrix} a_{n+1} \\ b_{n+1} \end{pmatrix} = \mathbf{A} \begin{pmatrix} a_n \\ b_n \end{pmatrix}$. Show how your answer is derived.
- (ii) [2 marks] Express the n -th power $\mathbf{A}^n$ of $\mathbf{A}$ as a product $\mathbf{BCB}^{-1}$ where $\mathbf{B}$ is a non-singular matrix and $\mathbf{C}$ is a diagonal matrix.
- (iii) [4 marks] Suppose the initial populations a_0 and b_0 in both towns are 100,000. Express each of a_n and b_n as an equation in terms of n . Show your workings clearly. (You may use MATLAB where possible.)
- (iv) [3 marks] Explain why the ratio of the populations of the two towns in the long run is always the same regardless of the initial populations a_0 and b_0 .

Q19 (13 marks)

Answer this question by writing in the exam answer booklet. Do not type in Exemplify directly. Indicate Q19 at the top on each page of the answer to this question. Indicate the part numbers on the left margin. You may use MATLAB for this question. But you should show how your answers are derived.

Consider the SDE $\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} -1 & 4 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$.

- (i) [2 marks] Write down a fundamental set of solutions of the SDE.
- (ii) [3 marks] Solve the initial value problem with $y_1(0) = 1$ and $y_2(0) = 0$. Show your working clearly.
- (iii) [2 marks] What is the type and stability of the equilibrium point of the SDE?
- (iv) [4 marks] Sketch a phase portrait of the SDE. Indicate in your diagram the trajectory that corresponds to the solution in (ii).
- (v) [2 marks] Write down a particular solution of the SDE that corresponds to a trajectory which moves closer and closer to the equilibrium point.